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POLITECNICO
MILANO 1863

Requirement Analysis and Specification Document

Deliverable:	RASD
Title:	Requirement Analysis and Verification Document
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Version:	1.0
Date:	28/10/2025
Download page:	https://github.com/angelo-cantoli/CantoliBedini
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1 Introduction

1.1 Purpose

BBP is a software designed to simplify the life of everyone who goes on bike trips and wants to enjoy them without worrying about finding the best routes.

It is intended as a community for cyclists, in which every person can share information about their personal routes, so that everyone may take advantage of the information and experience of other bikers. The application is intended for smartphone, where it can make use of GPS and sensor data to seamlessly reconstruct the path a cyclist has taken.

1.1.1 Goals

- G1: Cyclists maintain accurate and comprehensive records of their bike trips
 - G1.1: Bike trip data is automatically tracked while cycling
 - G1.2: Recording bike trips remains possible without automatic tracking
 - G1.3: The records of cyclists' bike trips are persistent and always accessible
 - G1.4: Cyclists' performance statistics are observable throughout their trip history
 - G1.5: Recorded bike trips include the condition of the roads on the bike path
 - G1.6: Hazards and obstacles on the bike path are identified and recorded
 - G1.7: Recorded bike trips include meteorological conditions when available
- G2: Cyclists make informed route selections
 - G2.1: All known paths that cover a route are observable
 - G2.2: The most suitable paths between two locations are distinguishable
 - G2.3: Path suitability is objectively evaluable based on road condition and route effectiveness
- G3: A community knowledge base of bike path information is maintained by cyclists
 - G3.1: Cyclists contribute bike path information by sharing their recorded trips
 - G3.2: Published bike path information is accessible to all users regardless of registration

1.2 Scope

1.2.1 World Phenomena

- WP1: User goes cycling along roads
- WP2: Roads contain potholes or other obstacles
- WP3: Road conditions vary over time
- WP4: Meteorological conditions change over time
- WP5: User needs to find routes between two locations
- WP6: User decides to record a bike trip
- WP7: Cyclists decide to share information with a community of other cyclists

1.2.2 Shared Phenomena

- SP1: User creates an account
- SP2: User logs into their account
- SP3: User logs out of their account
- SP4: User starts trip recording
- SP5: User stops trip recording
- SP6: System confirms trip has been saved
- SP7: System displays statistics regarding saved trips
- SP8: System retrieves meteorological information from an external service
- SP9: System displays trip information enriched with weather data
- SP10: User manually inserts street names and status of a bike path
- SP11: User manually inserts the presence of an obstacle on the path
- SP12: User enables the system to automatically acquire data from their mobile device
- SP13: System detects if the user is biking based on speed data
- SP14: Mobile device's GPS provides location coordinates for path reconstruction
- SP15: Mobile device's accelerometer provides motion data
- SP16: Mobile device's gyroscope provides orientation data
- SP17: User confirms automatically detected obstacles
- SP18: User corrects automatically detected obstacles
- SP19: User publishes bike trip information
- SP20: User specifies an origin and destination for path visualization
- SP21: System displays relevant path information
- SP22: System displays paths between two locations, ranked based on score
- SP23: User views their recorded trips
- SP24: User edits one of their recorded trips
- SP25: User deletes one of their recorded trips

1.3 Definitions, Acronyms, Abbreviations

1.4 Revision History

1.5 Reference Documents

1.6 Document Structure

2 Overall Description

2.1 Product Perspective

2.2 Product Functions

Sign up & Login

Users can access the app in two ways:

- Unregistered access: accessing the app without registering an account gives a user access to the path visualization interface. By inserting a start and end location, the application will show on a map all the paths between the two points, sorted based on their score (as described in detail in the 'Path Visualization' product function).
- Registered access: the app also gives users the option to register a personal account. It will ask the user to insert a username, e-mail and a password, along with some optional personal data (name, date of birth, profile picture). If the username or email is already taken by another user, the system will notify that a different username must be chosen. A verification email will be sent to the provided address with an OTP, which will be requested by the system before the registration can be submitted. Once the registration is complete, the user can login to their account, which gives them the ability to record, and eventually publish, their bike trips, in addition to the features of the unregistered access. The user can log out from their account at any time.

Recording of Trips

The main functionality of BBP is to allow registered users to record and save their bike trips. There are two ways to record a trip in BBP:

- Manual recording: if the user wants to record their trip manually, they must insert all relevant information into the BBP application: the start and end locations of their trip, the names of the streets present in the path (in the order in which they were traversed), their status (selected from predefined status values: optimal, medium, sufficient, requires maintenance), and the start and end times of the trip. Users can also manually mark the locations of obstacles encountered along the path. The system will reconstruct the path based on the streets provided and their order.
- Automatic recording: if the user wants to record their trip automatically, they must begin the automated recording before the start of their trip, and allow BBP to acquire data from their mobile device. BBP will then start tracking the GPS position of the user's mobile device to determine what path they are taking; the tracking will begin once the speed of the mobile device is estimated to be elevated enough to assume that they have started biking (e.g. 15 km/h), and will not end until the users manually halts the recording. BBP will simultaneously access some of the mobile device's sensors, such as the gyroscope or the accelerometer, so that if a notable movement is registered (due to the presence of some obstacle on the road, such as a pothole) an obstacle marker is added to that point in the path. The system will also determine the status of the road based on the number of obstacles and sensor data patterns. Once the trip has ended the user must stop the recording, at which point BBP will save the trip.

The recorded trips are saved in a database accessible only to the user; trips can be published at any time so that other users may visualize them, however if the trip data was recorded automatically BBP will first ask the user to verify that the collected information is accurate, and if it is not to manually correct it by removing incorrect obstacle markers and adjusting the status of the roads, before publishing the trip.

Trip Statistics Generation

Once a user has recorded a trip, the system will automatically generate some statistics about the trip. The statistics include:

- total distance covered
- average speed
- time taken to complete the trip
- total elevation difference between start and end points
- effective elevation traversed
- average slope of the path

These statistics are calculated based on the recorded path data and the inserted start/end times. The system will also query an external service to obtain meteorological data relative to the path at that time: if it is available, it will be added to the path statistics. The computed statistics are stored with the corresponding path in the database; the users have the option to access them when visualizing a path. Additionally, the system will compute aggregate statistics across all the trips the user has recorded (e.g. average time taken to complete trips, average elevation difference of trips, etc.)

Trip History Management

Users have the option to view all their past recorded trips. The trips are presented in a list format in chronological order, with the date of recording, the name and the start and end locations immediately visible. By selecting a specific trip, the user can view the computed statistics for that trip, and a map view of the path: there is an option to open the path in a third party navigation app (e.g. Google Maps), so that the user can navigate the same route again if they desire. The user has the option to edit or delete the trips at any time, and can also choose to publish the trip: if the data for the trip was automatically generated and the user has not yet reviewed it, the system will prompt the user to verify and correct the information before publishing. If a trip that has already been published is deleted or modified, the system will propagate the action to the published trip (e.g. if a user deletes a published trip from their personal trip history, the trip will no longer be available to other users); the user also has the option to make a published trip private again at any time, immediately removing it from public availability.

Path Visualization

Users, both registered and unregistered, can visualize published bike trips on a map. The application will prompt the user to insert a start and end location: once inserted, the system will search for all paths that connect those two points. The system then computes a score for each of the paths: the score is based on the status of the roads in the path, the time it would take for the user to reach the destination by following the path (calculated by considering the total length of the path and the elevation difference), the amount of obstacles along the path and the difficulty of the path (calculated by considering the average slope of the path). The system then shows the user all the identified paths on a map: the path with the highest score will be highlighted, while the others will be shown with lower opacity, and the scores will all be visible. Only the 5 paths with the highest score will be immediately visible, to avoid cluttering; the user will have the option to make all possible paths visible. The user can select a path by clicking on it: the application will then give the user the option to view the statistics of the path, select the path as the main highlighted path (if it isn't already), and to open the path in a third party navigation app (e.g. Google Maps). The user will also have the option to view all paths in a list, sorted by score.

2.3 User Characteristics

2.4 Assumptions, Dependencies and Constraints

2.4.1 Domain Assumptions

- D1: Each registered user has a valid email address

- D2: Users have a mobile device capable of running the BBP application
- D3: Users' mobile devices have GPS capabilities, with typical smartphone precision (5-10 meters)
- D4: GPS signals are generally available with sufficient precision while cycling
- D5: Registered users who use the automated recording functionality have accelerometer and gyroscope sensors in their mobile devices
- D6: Users' have internet access when using online features (e.g. path publication and visualization, weather data retrieval)
- D7: Users bike on roads intended for cycling (where bike tracks exist, or cars are rare and speed limits are compatible with biking)
- D8: Users who manually record trips bike on named roads which are marked on maps
- D9: A sufficient number of registered users are willing to publish their bike routes
- D10: Obstacles on bike paths cause a significant enough shift in movement to be detectable through the mobile device's sensors
- D11: Users can assess road surface conditions and categorize them in the predefined status values
- D12: Bike paths can be reconstructed reasonably accurately based on the names of the streets and the order in which they were traversed

3 Specific Requirements

3.1 External Interface Requirements

3.2 Functional Requirements

3.2.1 Mapping on requirements

3.2.2 Use case diagrams

3.2.3 Use cases

UC1. Unregistered user signs up

Actor	Unregistered user
Entry conditions	User is on login page of BBP
Event flow	1. User clicks to register on BBP 2. BBP asks user to insert his informations for the account 3. User inserts his informations 4. BBP validates the inserted personal informations 5. BBP sends a verification email to the provided address with an OTP 6. User inserts received OTP on BPP 7. BBP validates inserted OTP 8. BBP sends the registration outcome
Exit conditions	An account is created
Exceptions	Username already taken Password not valid Connection error while registering OTP not valid

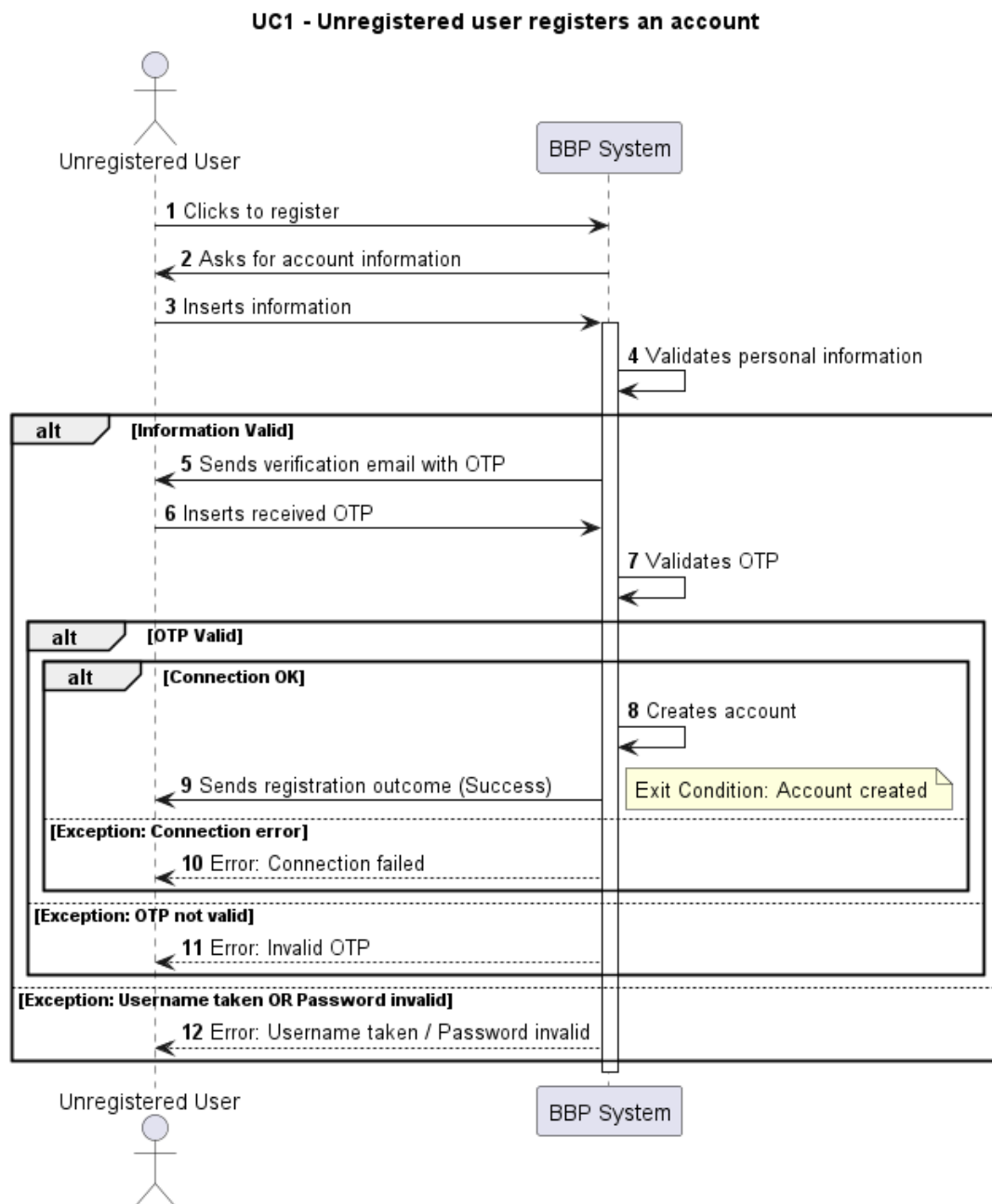


Figure 1: Use Case Diagram for UC1: Unregistered user signs up

UC2. User logs in the account

Actor	Registered user
Entry conditions	Unregistered is on login page of BBP
Event flow	1. User clicks to login on BBP 2. BBP asks user to insert username and password of the account he wants to log in 3. User inserts his username 4. BBP validates the inserted username and password 5. BBP sends login outcome
Exit conditions	User logs BBP system
Exceptions	Inserted username not existent Wrong inserted password for specified username Connection error while logging in

UC2 - User logs in the account

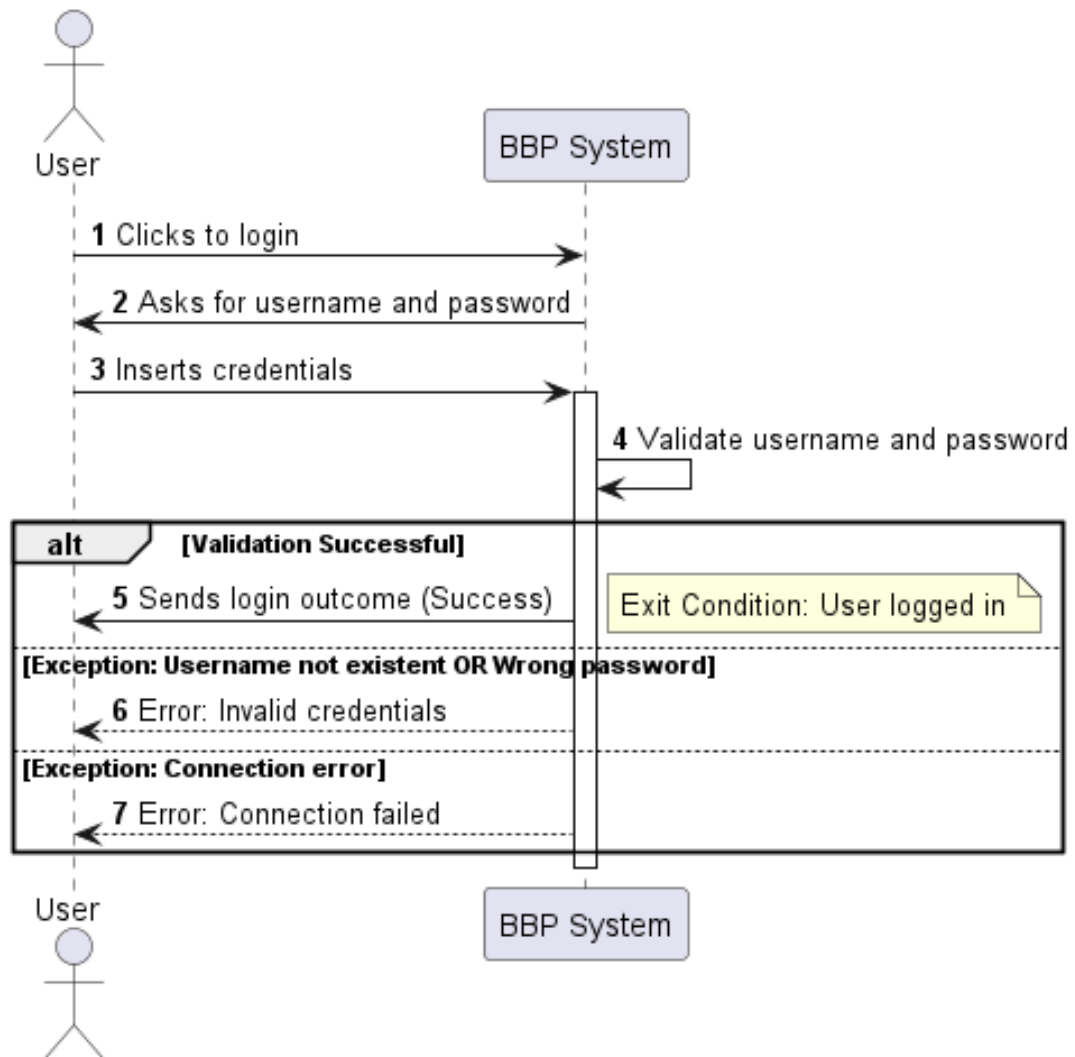


Figure 2: Use Case Diagram for UC2: User logs in the account

UC3. User manually adds bike trip

Actor	Registered user
Entry conditions	User is registered and logged in User is on "New trip" page
Event flow	1. User clicks on "Insert a bike trip" button 2. BBP asks to insert a name for the trip 3. User inserts name of the trip 4. BBP validates inserted name 5. User inserts traveled roads 6. BBP calculate and returns the complete traveled route 7. User confirms calculated route 8. User inserted optional information, i.e obstacles 9. BBP calculates status based on inserted informations and shows to user 10. User clicks "Save trip" button 11. BBP saves trip, and informs user of outcome
Exit conditions	Inserted trip is saved
Exceptions	BBP can't calculate a route based on inserted roads User already saved a route with the same name

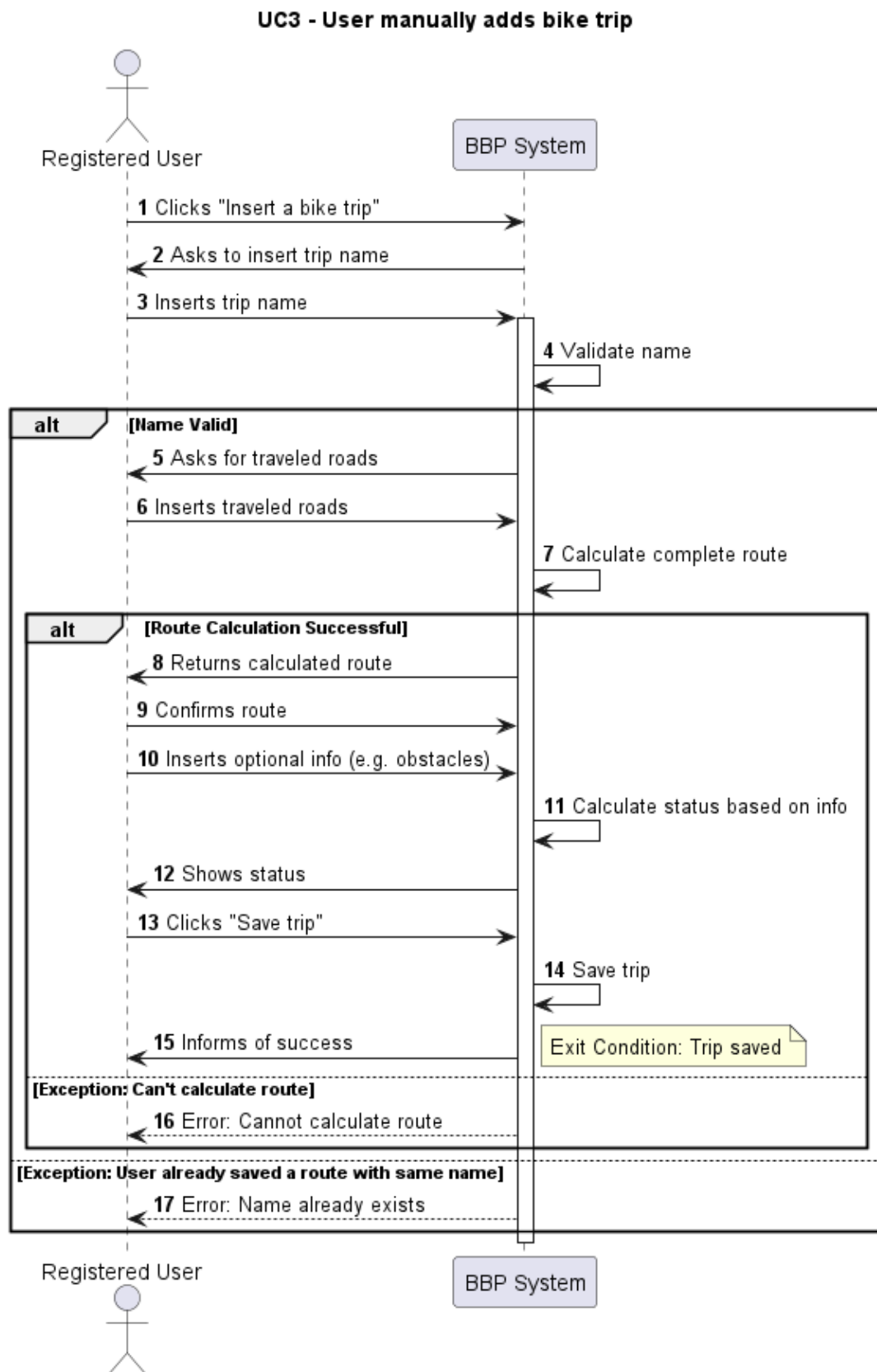


Figure 3: Use Case Diagram for UC3: User manually adds bike trip

UC4. User automatically records bike trip

Actor	Registered user
Entry conditions	User is registered and logged in User is on "New trip" page
Event flow	1. User clicks to start automatic recording mode 2. BBP starts taking data from user's phone GPS 3. BBP recognize user is cycling, based on his velocity from sensors 4. BBP traces position and encountered obstacles with user's phone sensors and GPS 5. BBP recognize user stopped cycling 6. User finished automatic recording 7. BBP asks user to confirm or correct recorded data 8. User optionally corrects data and click confirm 9. BBP calculate status based on recorded and inserted data and shows to user 10. User insert a name for the trip and clicks "Save trip" button 11. BBP saves trip, and informs user of outcome
Exit conditions	Recorded trip is saved
Exceptions	GPS signal lost Sensor permissions negated User already saved a route with the same name User discards recorded trip

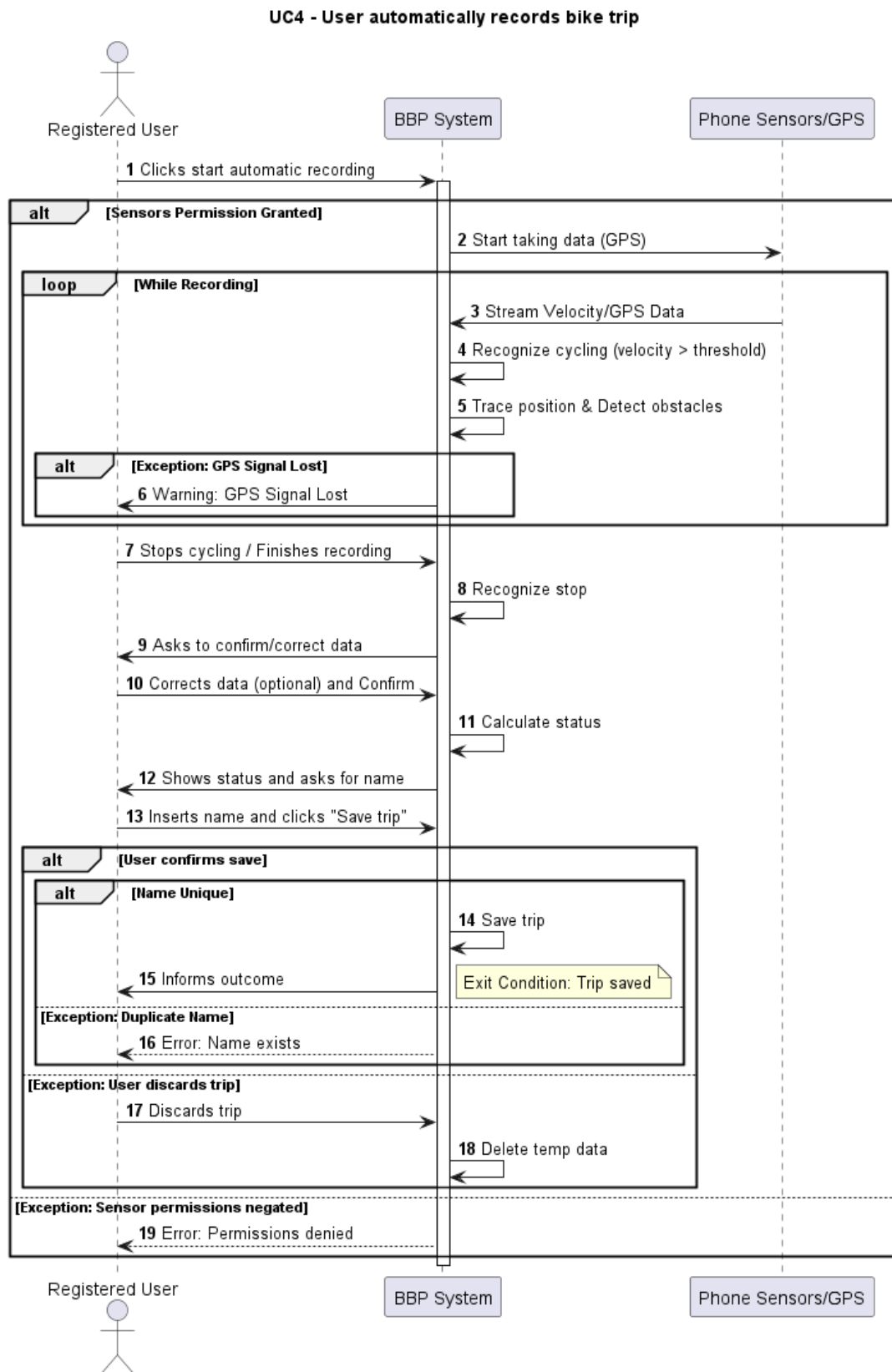


Figure 4: Use Case Diagram for UC4: User automatically records bike trip

UC5. User views his bike trip history

Actor	Registered user
Entry conditions	User is registered and logged in User is on "Your trips" page
Event flow	1. User opens on "Bike trip history" section 2. BBP loads history of trips of user 3. User visualizes his trips
Exit conditions	User closes "Bike trip history" section
Exceptions	User does not have any saved trips

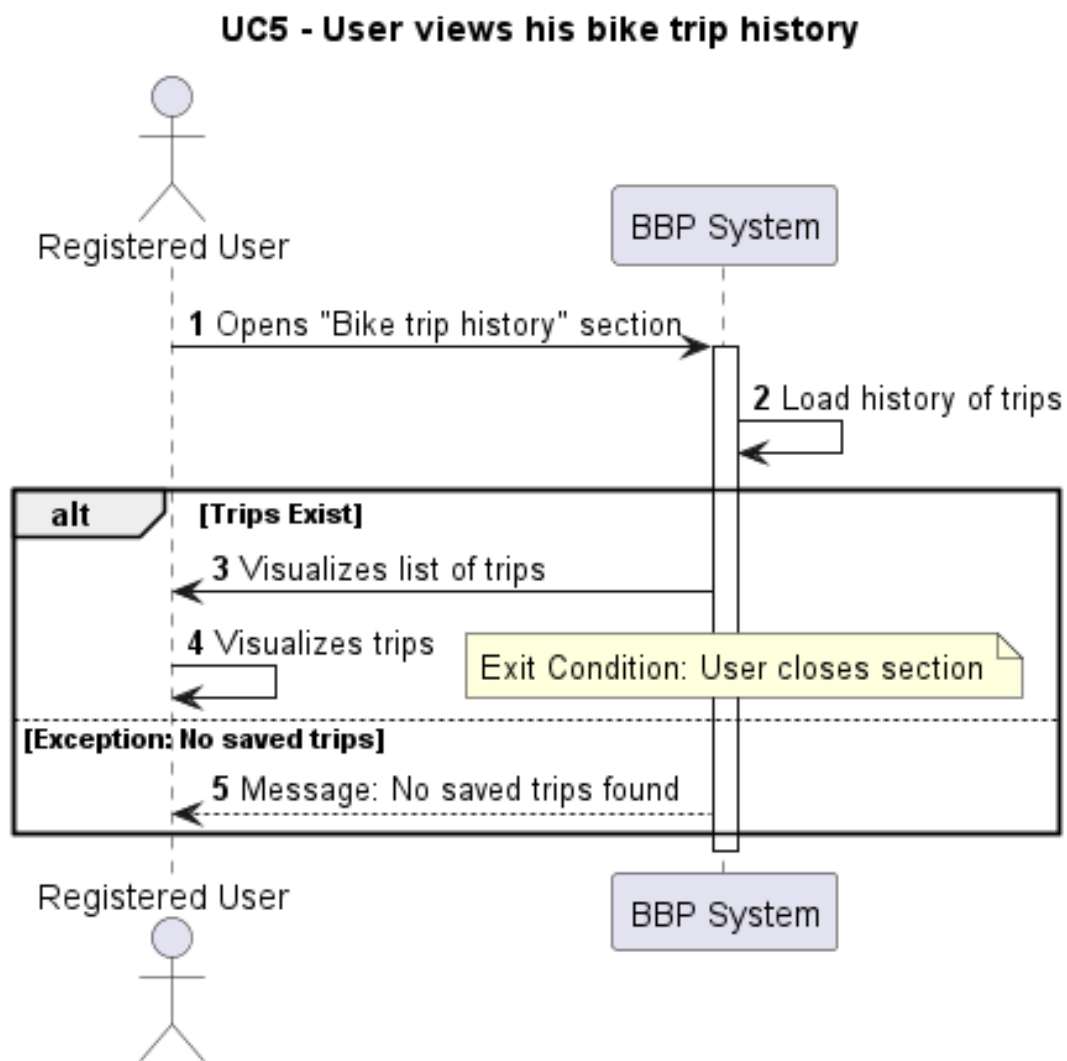


Figure 5: Use Case Diagram for UC5: User views his bike trip history

UC6. User views his bike trip statistics

Actor	Registered user
Entry conditions	User is registered and logged in User selected a specific trip from history
Event flow	1. User clicks on "View statistics" button 2. BBP loads complessive statistics of traveled bike trips, and statistics for each trip 3. User clicks on a specific trip 4. BBP visualizes in-depth statistics for that trip
Exit conditions	User closes "Trips Statistics" section
Exceptions	User does not have any trip to calculate statistics on

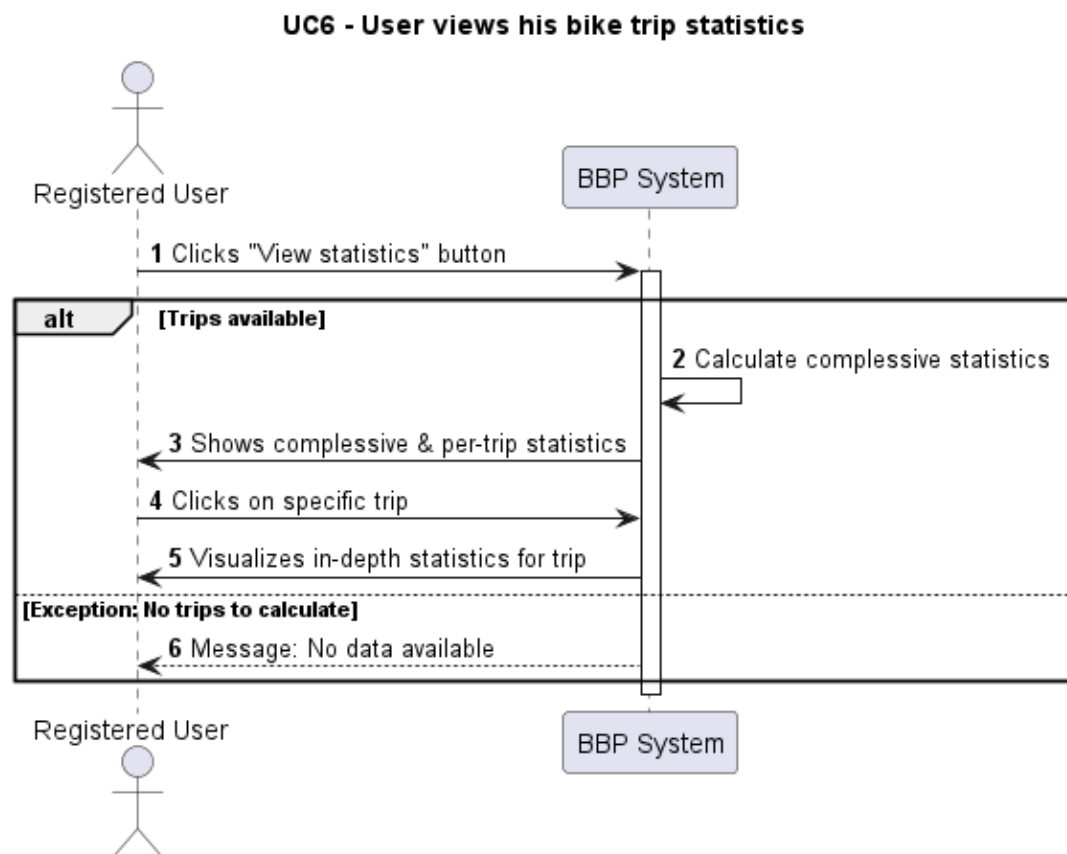


Figure 6: Use Case Diagram for UC6: User views his bike trip statistics

UC7. User publishes recorded trip

Actor	Registered user
Entry conditions	User is registered and logged in User selected a specific trip from history
Event flow	1. User click on "Publish trip" 2. BBP ask confirm to user if information are correct and if he is sure to make the trip public 3. User confirms the intention to publish 4. BBP publishes the trip, making it visible to all users
Exit conditions	User's trip is published and visible by all users
Exceptions	Connection error while publishing

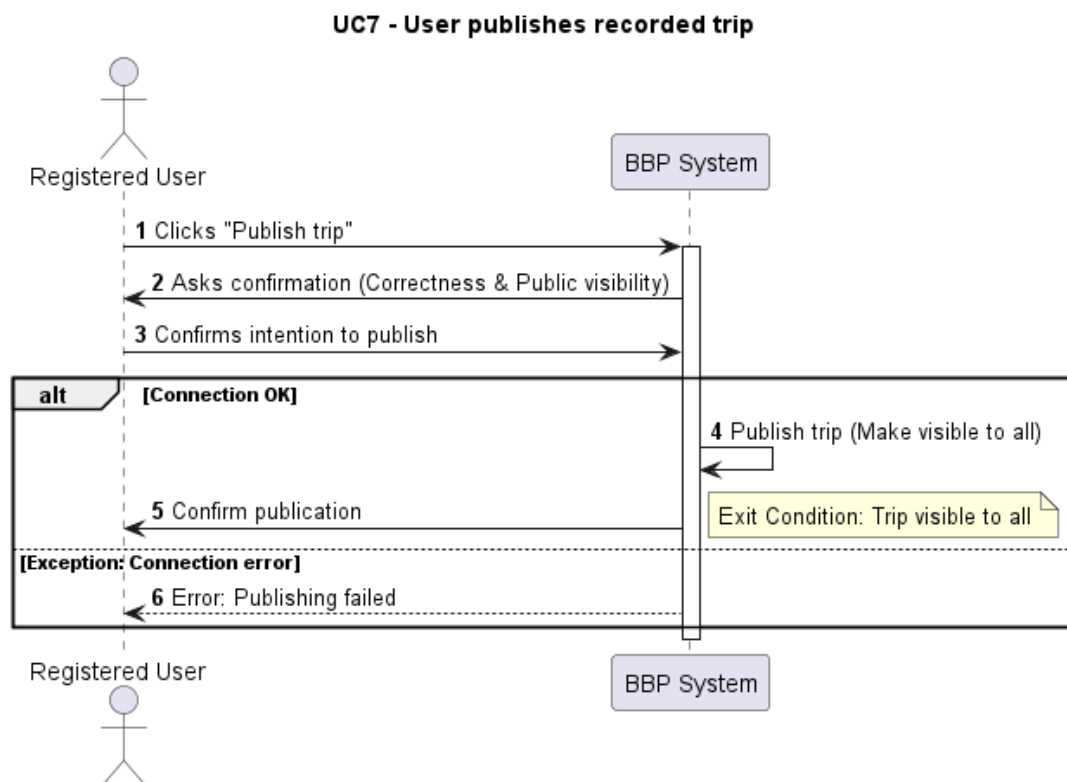


Figure 7: Use Case Diagram for UC7: User publishes recorded trip

UC8. User modify one of his saved trips

Actor	Registered user
Entry conditions	User is registered and logged in User selected a specific trip from history
Event flow	1. User click on "Modify trip" 2. BBP allows users to modify saved data about trip 3. User modifies trip data 4. BBP validate changes of trip data 5. User confirms the changes of trip data 6 BBP recomputes trip statistics based on modified data 7. BBP modifies trip data on database 8. If the trip is public BBP propagates changes for every user
Exit conditions	Trip data is modified
Exceptions	Inserted data is not valid Connection error while propagating changes of public trip

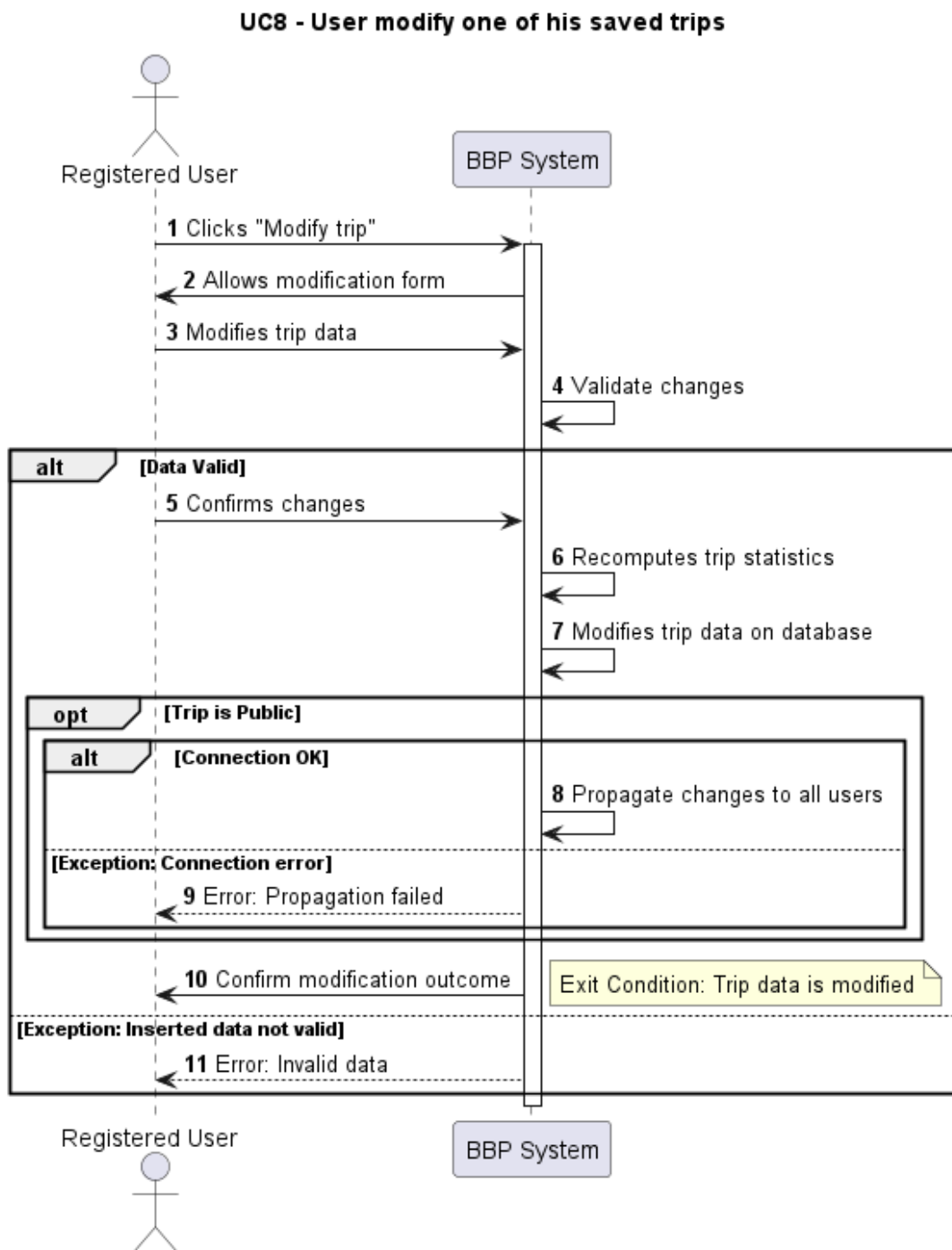


Figure 8: Use Case Diagram for UC8: User modifies one of his saved trips

UC9. User views possible bike routes on map

Actor	Registered user Unregistered user
Entry conditions	User is on "Path search" section
Event flow	1. BBP asks user to insert start and end point of route 2. User inserts start and end point 3. BBP visualizes every routes, with related score, between selected points
Exit conditions	Paths are visualized
Exceptions	Mapping service unavailable Connection error while retrieving paths No route found between the selected points

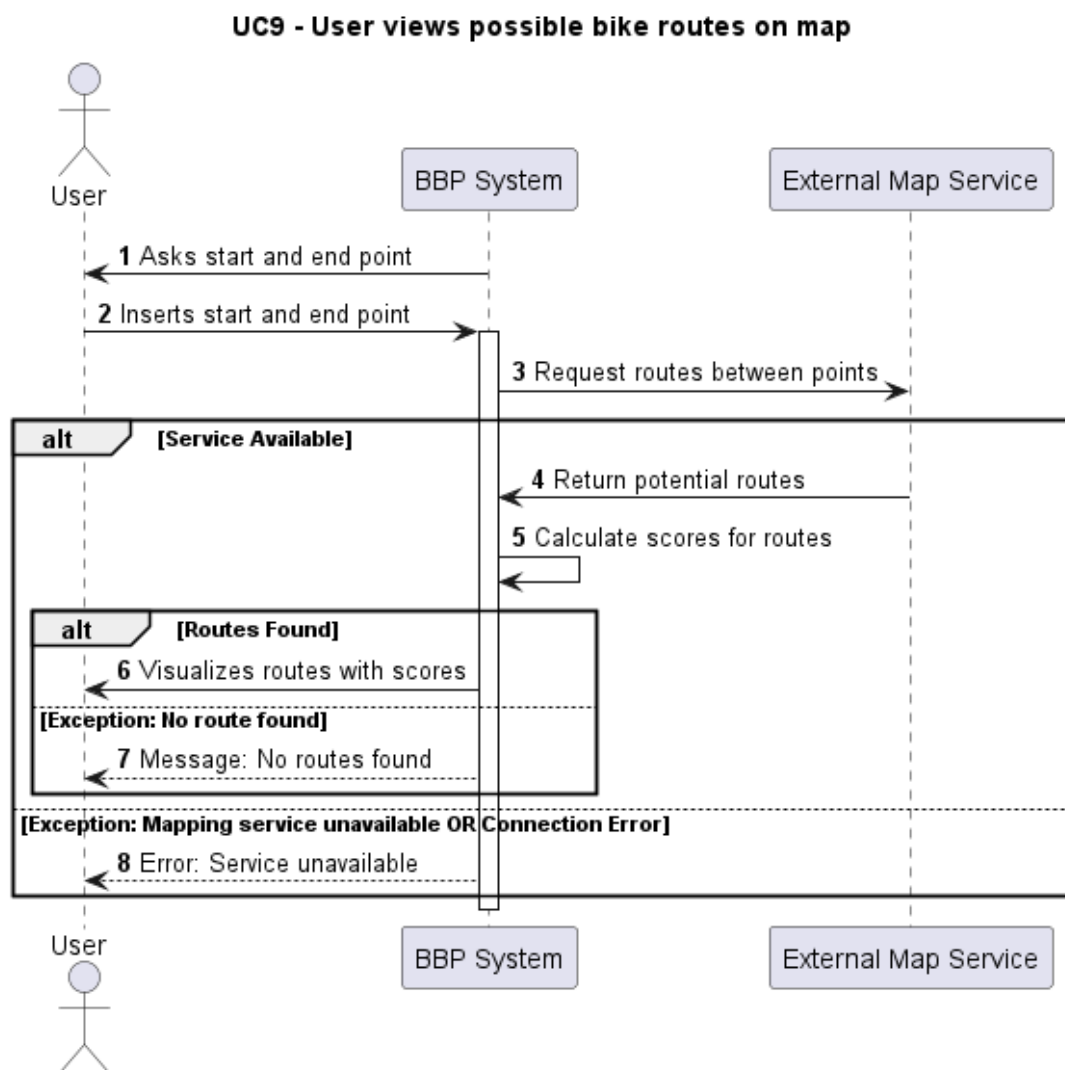


Figure 9: Use Case Diagram for UC9: User views possible bike routes on map

UC10. User views additional information about selected path on map

Actor	Registered user Unregistered user
Entry conditions	User is on "Path search" section visualizing paths between 2 points
Event flow	1. User clicks on a specific path on the map 2. BBP shows details about trip, i.e statistics, distance, obstacles etc..
Exit conditions	Additional informations of a trip are visualized
Exceptions	

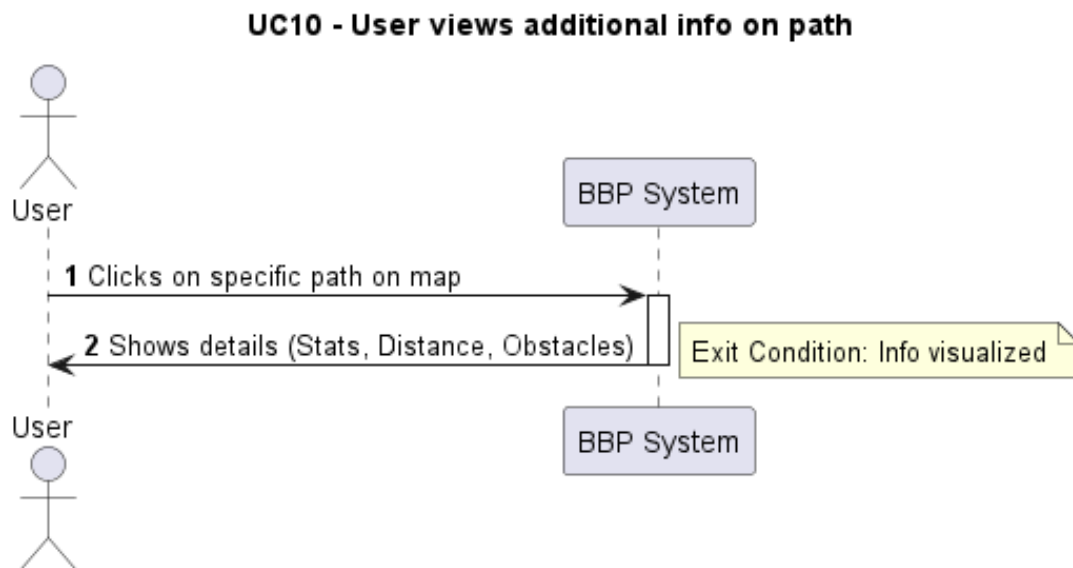


Figure 10: Use Case Diagram for UC10: User views additional information about selected path on map

UC11. User start navigation on selected bike path

Actor	Registered user Unregistered user
Entry conditions	User selected a path on the map
Event flow	1. User click to open route on a third-party mapping service 2. BBP opens third-party mapping service, giving it coordinates of start and end point
Exit conditions	Additional informations of a trip are visualized
Exceptions	third-party mapping service not available

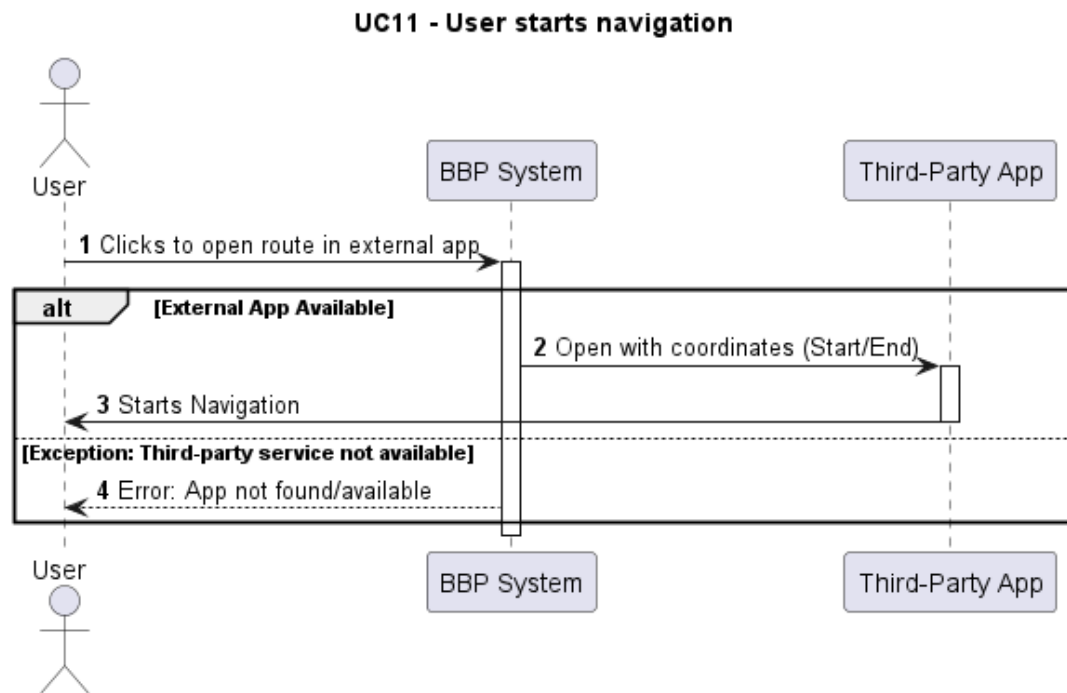


Figure 11: Use Case Diagram for UC11: User start navigation on selected bike path

UC12. User modifies profile data

Actor	Registered user
Entry conditions	User is registered and logged in User is on profile page of BPP
Event flow	1. User clicks on modify profile data 2. User insert new personal data 3. BBP validates new personal data 4. User confirms changes on personal data 5. BBP update user personal data
Exit conditions	User personal data are modified
Exceptions	New username already taken New password not valid Connection error while modifying profile data

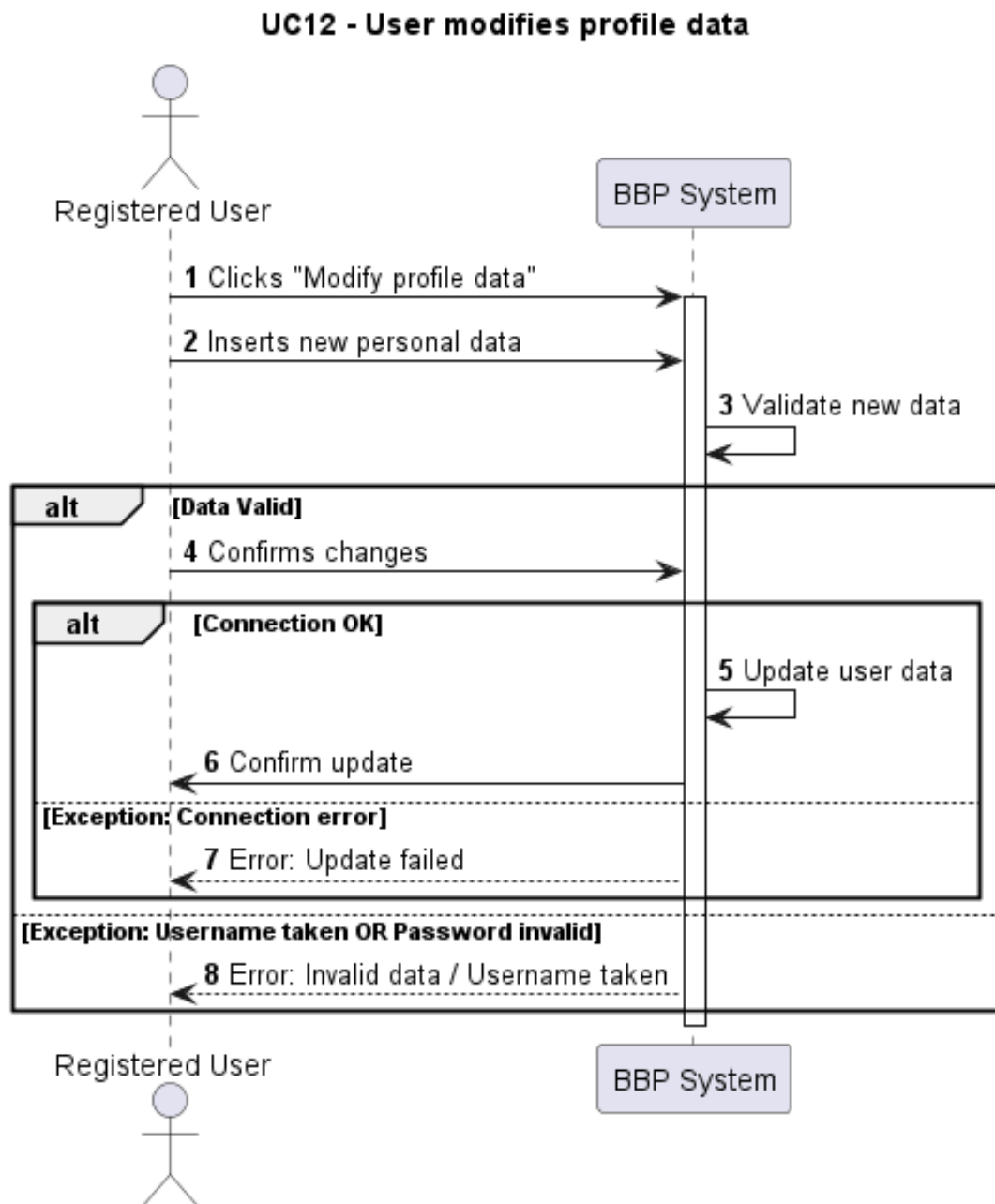


Figure 12: Use Case Diagram for UC12: User modifies profile data

UC13. User logs out the account

Actor	Registered user
Entry conditions	User is registered and logged in User is on profile page of BPP
Event flow	1. User clicks on log out button 2. BBP terminate access of user on the account 3. BBP returns user to log in page
Exit conditions	User is disconnected from the account
Exceptions	

UC13 - User logs out

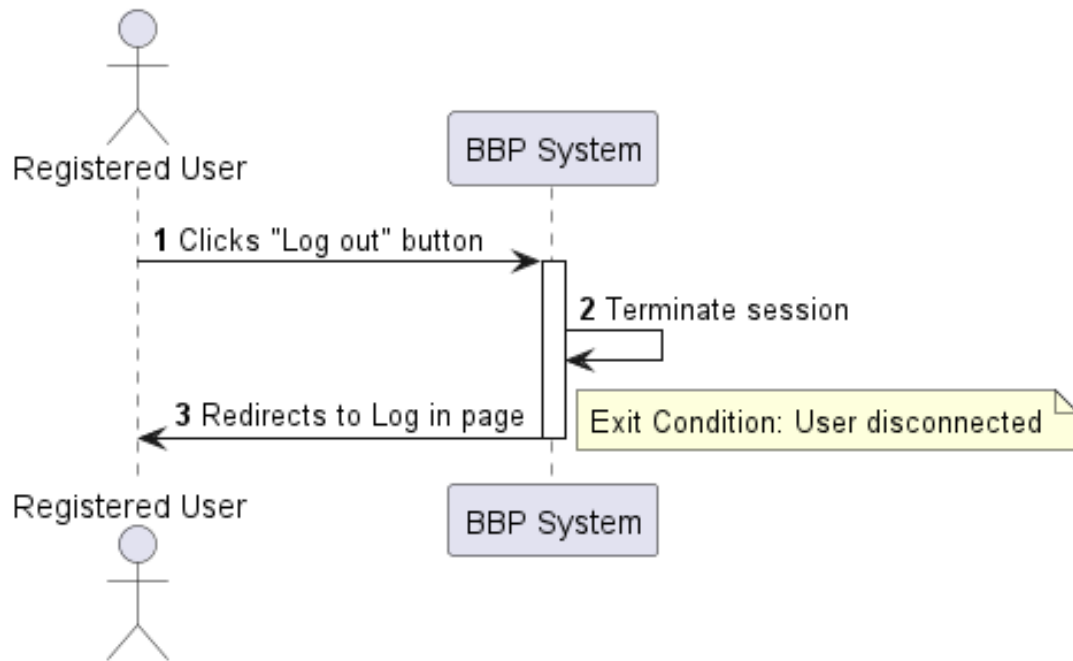


Figure 13: Use Case Diagram for UC13: User logs out the account

UC14. User deletes the account

Actor	Registered user
Entry conditions	User is registered and logged in User is on profile page of BPP
Event flow	1. User clicks on delete account button 2. BBP asks for confirmation 3. User confirms account deletion 4. BBP deletes user account and all associated data 5. BBP returns user to home page
Exit conditions	User is disconnected from the account and account is deleted
Exceptions	Connection error while deleting account

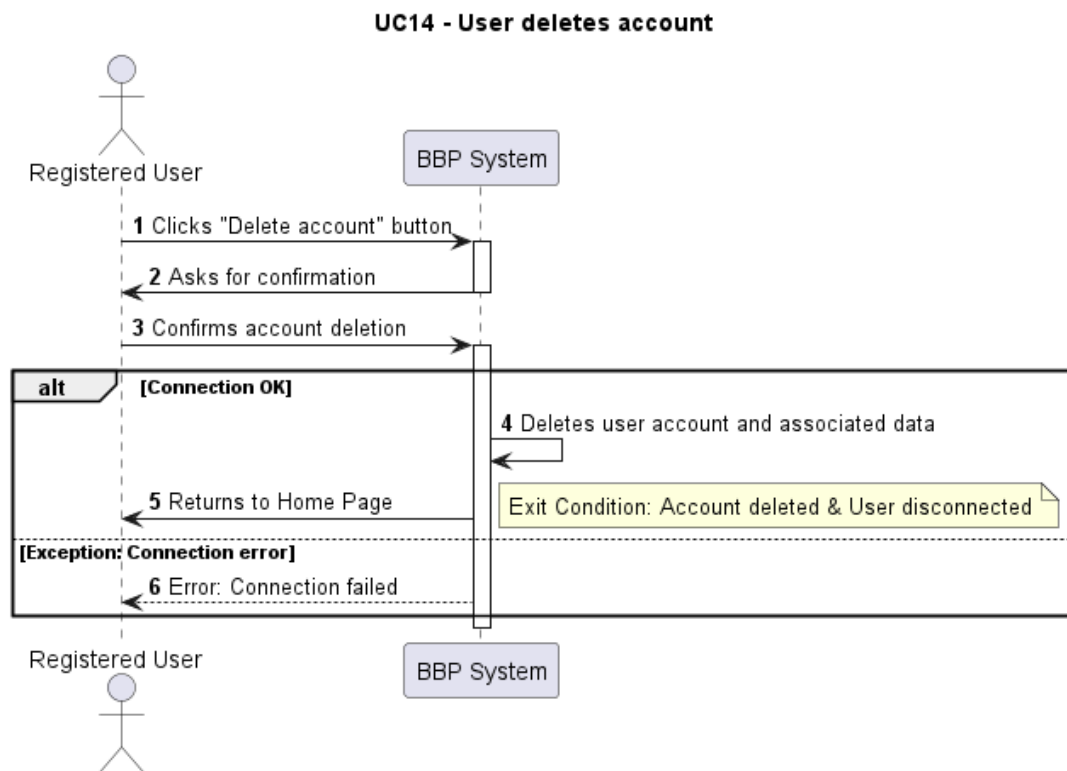


Figure 14: Use Case Diagram for UC14: User deletes the account

UC15. User recovers password of the account

Actor	Registered user
Entry conditions	User is on login page of BBP
Event flow	1. User clicks on "Forgotten password" button 2. BBP asks for email of the account 3. User inserts email associated to the account 4. BBP sends an OTP to the inserted email 5. User inserts received OTP on BPP 6. BBP validates inserted OTP 7. User inserts new password 8. BBP validates new password 9. User confirms new password 10. BBP updates password of the account 11. BBP returns user to login page
Exit conditions	Password is successfully reset
Exceptions	Connection error while resetting password Invalid OTP New password not valid

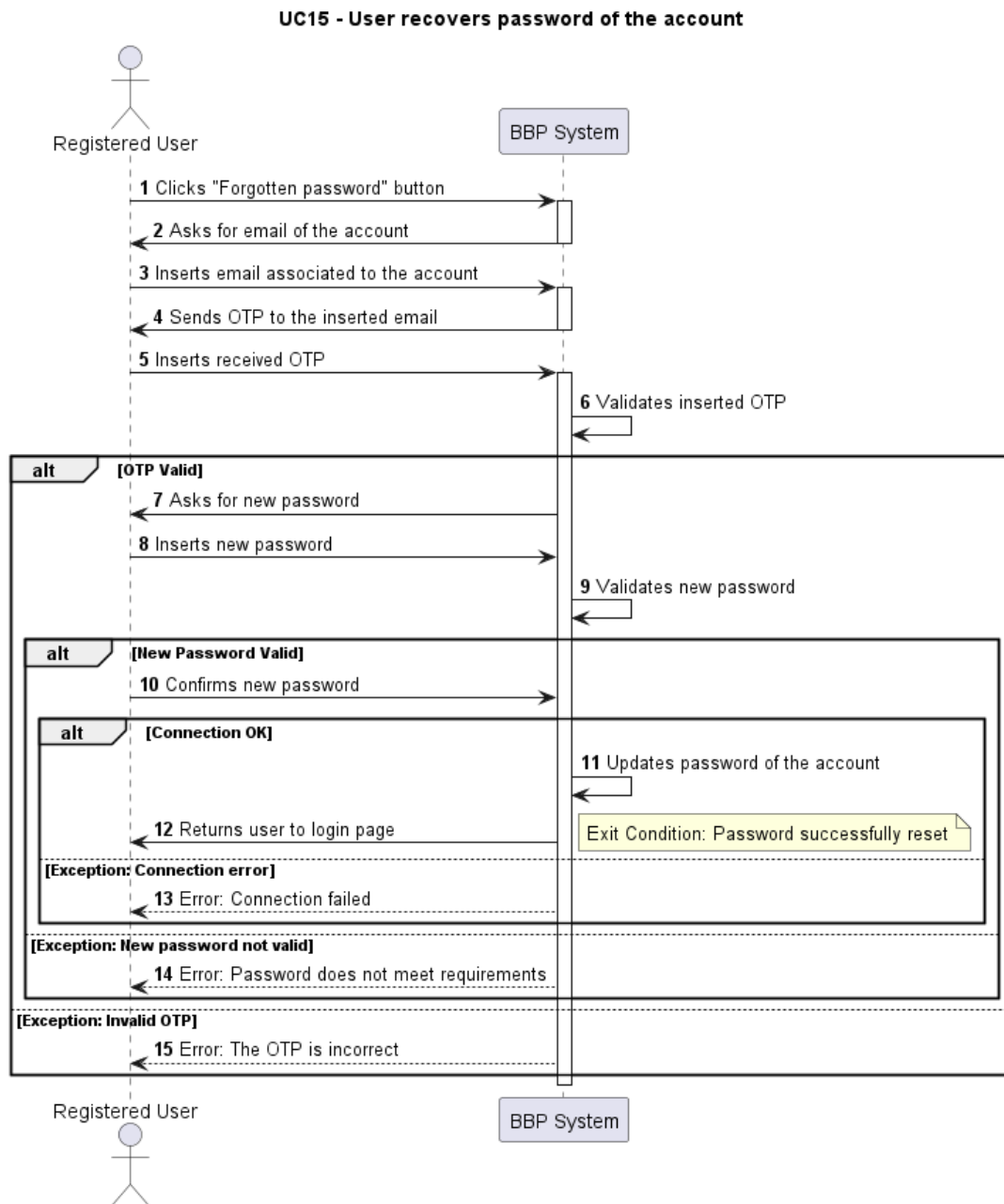


Figure 15: Use Case Diagram for UC15: User recovers password of the account

4 Formal Analysis Using Alloy

5 Effort Spent

References